

experience to their students. Most institutes initially train their students on simulators as handling of drones is an unwarranted cost.

Aerial mapping

In the commercial space, a lot of sectors such as mining, real estate, infrastructure, agriculture and oil have been relying on drones for data that is otherwise difficult to access. As per a study by Allied Market Research, aerial photography via drones is estimated to reach the US\$ 2.8 million mark worldwide by 2022. For instance, Mumbai based drone startup Airpix specialises in using drones for aerial mapping of areas that are otherwise difficult to reach. "We provide data acquisition and analysis solutions using UAVs for multiple industries. The solutions and reports provided by the company help its clients in project design, progress monitoring, volume calculations and inspection of critical assets," says Shinil Shekhar, co-founder of Airpix.

Shekhar adds, "Drones are just a platform. What you make of them depends on how you select the sensors based on your project requirement and what kind of insights you derive after processing that data." Currently, the company's client portfolio includes the Indian government and the Indian Railways as well as a few private firms such as Larsen & Toubro, Shapoorji & Pallonji and JSW Steel.

While globally, industrial use of drones for mapping is done extensively, India is still at a nascent stage. Industry players hold the current regulations responsible for the slow growth in this business category. "For drones to become a great business, healthy regulations have to be in place—regulations that will promote the technology, safeguard national security and ensure public safety," notes Shekhar.

Aerial cinematography and photography

This is one category that took flight immediately after drones made their

debut commercially. Drones are being extensively used in filming, news reporting, real-estate and wildlife photography. In India and the rest of the world, high-profile events such as weddings have been using drones to capture aerial views of the event for quite some time. And with Federal Aviation Administration (FAA) labelling the use of commercial drones last year, drone photography and cinematography are anticipated to grow.

Currently, drone photography is being pursued on an amateur level. But with the guidelines in place, this category is expected to gain a lot of attention from industries, as photography is pivotal to aerial mapping as well.

At present, the initial cost of proper drone photography is less than US\$ 10,000.

Dronevertising

Marketers across the world are always looking at connecting with their consumers innovatively. And since drones are relatively a new technology in the commercial space, brands are looking at ways to explore it. In fact, the advent of social media has made it mandatory for brands to try something new.

Interestingly, a few marketing agencies have been mushrooming across the world that specialise in drone advertising, which is popularly called dronevertising. DroneCast, Hoovy and Intel are three major players in the segment. Big shots such as PepsiCo, Coca-Cola and GE have already used drones in their brand campaigns to connect better with their consumers. The initial investment is somewhere between US\$ 20,000 and US\$ 25,000.

The mindset of marketers in different regions is a huge challenge. While



UAV training in action (Image courtesy: Maavan Aeronautics Pvt Ltd)

brands in the West are more experimental and are prepared to shell out more on a new technology, the same cannot be said for brands in India.

However, with the technology expected to become cheaper in the future, it may not be too far-fetched to say that drones would be the next logical step after digital media, and dronevertising may be the biggest trend in marketing in the future in India.

Drones for delivery

We cannot ignore the most basic functions of a drone—delivery. Amazon is developing a fleet of drones for its airborne service, Amazon Prime Air Delivery. The e-retailer's drone delivery service has successfully delivered goods to two of its customers in the UK. And while it is still in its nascent stage—drones can only deliver goods that weigh less than 2kg and need a helipad to land—we can expect drone manufacturers to come up with new user-friendly designs that will soon make deliveries easier.

Since India along with the rest of the world is moving towards digitalisation, and it is one of the fastest growing online retail markets, cheap airborne delivery may just be the next idea waiting to be commercialised.

Cost of infrastructure for drone delivery is a major challenge as companies in this space have to invest in additional infrastructure support such as helipads. However, drone delivery is expected to be a core part of the logistics sector in the future. **EFY**

—Purba Das, senior business journalist, EFY